

Literature-Implied Answer To The Amenable Calkin Algebra Question In arXiv:0909.2628

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Status

literature_implied_answer (full problem). This packet records a positive answer to the Calkin-algebra question in Runde [1], obtained by direct identification with the $C(K)$ -Calkin realization theorem of Motakis–Puglisi–Zisimopoulou [2]. The supporting authors do not explicitly say that they are answering Runde’s question; the implication is agent-identified.

Source Question

In the final section, ‘Questions’, Runde asks:

Is there a Banach space E such that $\mathcal{C}(E)$ is amenable and infinite-dimensional?

Here $\mathcal{C}(E) = \mathcal{B}(E)/\mathcal{K}(E)$ is the Calkin algebra.

Supporting Theorem

Motakis–Puglisi–Zisimopoulou prove, as Theorem B in the introduction and as Theorem 5.1 in the final section, that for every countable compact metric space K there is a Banach space X , indeed an $\mathcal{L}_\infty$ -space with $X^* \cong \ell_1$, such that

$$\mathcal{C}(X) \cong C(K)$$

as Banach algebras.

Identification

Let

$$K = \{0\} \cup \{1/n : n \in \mathbb{N}\} \subset \mathbb{R}.$$

Then K is an infinite countable compact metric space. By [2], there exists a Banach space X such that $\mathcal{C}(X) \cong C(K)$ as Banach algebras. The algebra $C(K)$ is infinite-dimensional, since K is infinite and compact Hausdorff.

Moreover $C(K)$ is a commutative C^* -algebra. Commutative C^* -algebras are nuclear, and Runde’s source paper recalls the standard theorem that nuclear C^* -algebras are amenable as Banach algebras. Hence $C(K)$ is amenable. Amenability is preserved under Banach algebra isomorphism, so $\mathcal{C}(X)$ is amenable and infinite-dimensional.

Scope And Search

This is not a new construction from the run. It is a full positive literature-implied answer to Runde's question. Cheap run indexes were searched for '0909.2628', 'amenable Calkin', 'C(K) Calkin', and close variants; no exact prior packet was found. Local source search and web search both surfaced arXiv:1407.8073 as the decisive supporting source.

References

- [1] V. Runde, *(Non-)amenability of $\mathcal{B}(E)$* , arXiv:0909.2628, 2009.
- [2] P. Motakis, D. Puglisi, and D. Zisimopoulou, *A hierarchy of Banach spaces with C(K) Calkin Algebras*, arXiv:1407.8073, 2014.